Series Sum

```
#include <iostream>
using namespace std;

int main() {
   int choice, n, start, end;
   int sum = 0;

   cout << "1. Sum of n natural numbers";
   cout << "\n2. Sum of squares of n natural numbers";
   cout << "\n3. Sum of numbers in a range";
   cout << "\n4. Sum of squares in a range";

   cout << "\nEnter choice: ";
   cin >> choice;

   switch (choice) {
      case 1:
         cout << "Enter n: ";
         cin >> n;

         sum = n * (n + 1) / 2;
         cout << "Sum = " << sum;
         break;

      case 2:
         cout << "Enter n: ";
         cin >> n;

         sum = n * (n + 1) * (2 * n + 1) / 6;
```

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            cout << "Sum of squares = " << sum;
            break;

        case 3:
            cout << "Enter start and end: ";
            cin >> start >> end;

            for (int i = start; i <= end; i++)
                sum += i;

            cout << "Sum = " << sum;
            break;

        case 4:
            cout << "Enter start and end: ";
            cin >> start >> end;

            for (int i = start; i <= end; i++)
                sum += i * i;

            cout << "Sum of squares = " << sum;
            break;

        default:
            cout << "Invalid choice";
    }

    return 0;
}
```